

Smart Traffic Lights Using Local AI

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Abstract—Traffic congestion can change significantly during different periods of the day and at different road intersections. Conventional traffic signals commonly use predefined timing plans, which may not respond effectively to changing traffic conditions. Artificial intelligence and machine vision provide methods for obtaining traffic information and supporting adaptive traffic signal control. Previous research has studied learning-based traffic signal control and machine-vision-based traffic monitoring [1], [2].

This paper presents a conceptual architecture for a smart traffic-light system using local AI processing. In the proposed architecture, cameras collect traffic information and a local computing unit processes the information near the intersection. The resulting decision can be sent to the traffic-light controller. The architecture also considers emergency vehicle priority as a possible extension. The paper describes the system design, processing flow, possible advantages, limitations, and future implementation requirements.

Index Terms—Smart traffic lights, local AI, machine vision, adaptive traffic signal control, traffic management, emergency vehicle priority.

I. INTRODUCTION

Traffic congestion is a common challenge in urban transportation systems. Traffic signals are used to control the movement of vehicles at road intersections. Traditional traffic signals can operate using fixed or predefined timing plans. Although such systems are simple to implement, predefined timings may not respond well when traffic conditions change.

Artificial intelligence can be used to analyze traffic-related information and support traffic signal control. Research on deep reinforcement learning has investigated different approaches for traffic signal control and identified several research challenges in the field [1].

Machine vision is another approach for obtaining information from road traffic. Camera-based systems can be used to identify vehicles and estimate traffic-flow characteristics. Gorodokin et al. investigated machine-vision-based adaptive traffic-light control and considered traffic-flow information for adjusting traffic-light operation [2].

This paper proposes a conceptual architecture that combines camera-based traffic observation with local AI processing. The main idea is to process traffic information close to the intersection and use the resulting information to support traffic-light decisions.

II. RELATED WORK

Research in intelligent traffic signal control includes machine-learning, reinforcement-learning, and machine-vision approaches.

Rasheed et al. presented a review of deep reinforcement learning for traffic signal control. Their review discusses different deep reinforcement learning architectures, traffic-control attributes, simulation platforms, and open research problems [1].

Gorodokin et al. studied adaptive traffic-light control using machine vision. Their work considered the use of traffic cameras and traffic-flow parameters to support adaptive control of traffic-light phases and cycle duration [2].

These studies show that traffic information can be used to support adaptive traffic signal control. The present paper focuses on a conceptual local-processing architecture in which traffic information can be analyzed close to the road intersection.

III. PROBLEM STATEMENT

A traffic intersection can experience different traffic conditions during different periods. A fixed timing plan may not always match the current traffic demand.

The problem considered in this paper is how traffic information can be collected and processed locally so that a traffic-light controller can use current traffic conditions when making a signal decision.

An additional application considered by the proposed architecture is emergency vehicle priority. If an emergency vehicle is identified near an intersection, the system could provide a mechanism for modifying the signal operation. Such a mechanism would require reliable detection, coordination, and safety validation before real-world deployment.

IV. OBJECTIVES

The objectives of the proposed system are:

- To collect traffic information using road-side cameras.
- To process traffic information using a local computing unit.
- To support adaptive traffic-light decisions.
- To reduce dependence on a remote server for every real-time processing step.
- To investigate emergency vehicle priority as a possible application.

- To develop a conceptual architecture that can be tested in future work.

V. PROPOSED SYSTEM

The proposed system consists of four main components:

- 1) Street Camera
- 2) Local AI Processing Unit
- 3) Optional Cloud Server
- 4) Traffic-Light Controller

The street camera collects information about vehicles and traffic conditions. The local processing unit analyzes the captured information. Based on the analyzed information, a traffic signal decision can be generated.

The cloud server is considered optional. It can be used for storage, monitoring, model management, or further analysis. However, the conceptual architecture does not require the cloud server for every real-time traffic decision.

The traffic-light controller receives the decision generated by the processing system and applies the appropriate signal operation.

VI. SYSTEM ARCHITECTURE

The basic information flow of the proposed architecture is:

Street Camera → Local AI Unit → Traffic Controller

An optional cloud connection can be included:

Local AI Unit → Cloud Server

The cloud server may be used for storing selected information or performing additional analysis.

VII. SYSTEM FLOWCHART

Figure 1 shows the conceptual processing flow.

VIII. TRAFFIC STATE REPRESENTATION

The information obtained from the camera can be represented as a traffic-state vector:

$$S = [N \quad Q \quad V \quad E] \quad (1)$$

where:

- N represents the number of detected vehicles.
- Q represents queue-related information.
- V represents estimated traffic-speed information.
- E represents the emergency-vehicle detection status.

The signal action can be represented generally as:

$$A = f(S) \quad (2)$$

where S is the observed traffic state and A is the selected traffic-light action.

For the conceptual emergency-priority logic:

$$A = \begin{cases} \text{Emergency Priority,} & E = 1, \\ \text{Adaptive Control,} & E = 0. \end{cases} \quad (3)$$

These equations describe the proposed concept. They are not experimental results.

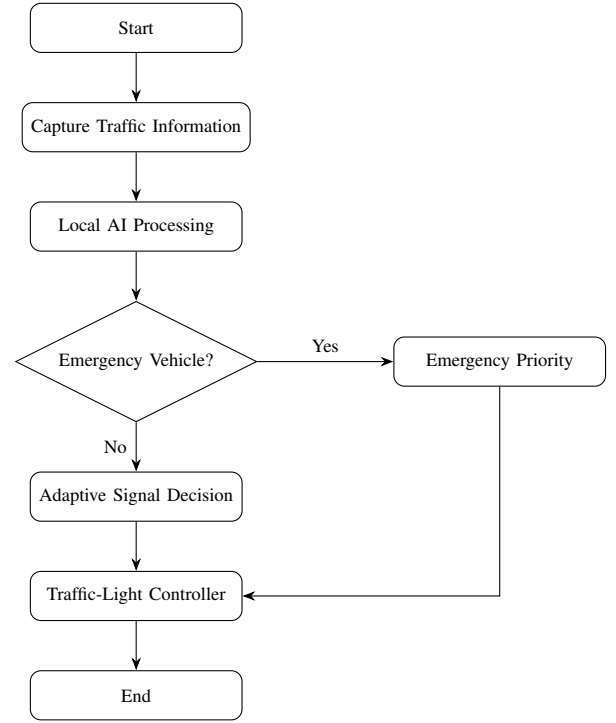


Fig. 1. Conceptual flowchart of the proposed smart traffic-light system.

IX. QUALITATIVE COMPARISON

The approaches can be compared based on their general characteristics rather than using unsupported numerical performance values.

TABLE I
QUALITATIVE COMPARISON

Feature	Fixed Timing	Cloud AI	Local AI
Traffic data usage	Low	High	High
Local processing	No	Limited	Yes
Adaptive control	Limited	Possible	Possible
Internet dependency	Low	Higher	Lower for local decisions
Emergency priority	Not inherent	Possible	Possible

The table is a conceptual comparison and does not represent measured experimental performance.

X. RESEARCH GAP

Previous studies have investigated adaptive traffic signal control using machine vision and learning-based approaches [1], [2].

Based on the literature considered in this sample paper, the proposed work focuses on the integration of local traffic processing and emergency vehicle priority within a single conceptual architecture.

The proposed system is therefore presented as a design concept, not as a completed experimental system. Future

research is required to implement the system and evaluate its performance under controlled conditions.

XI. ADVANTAGES

The proposed architecture has several potential advantages:

- Traffic information can be processed near the intersection.
- The system can use current traffic information rather than only predefined timing.
- Local processing can reduce the need to send all raw camera data to a remote server.
- Emergency vehicle priority can be included as a control condition.
- The architecture can be extended with different AI or machine-vision models.

These are potential advantages of the proposed architecture and are not experimentally measured results.

XII. LIMITATIONS

The proposed system has several limitations.

First, reliable traffic detection depends on camera quality, visibility, weather conditions, and the selected detection method.

Second, emergency vehicle detection must be reliable because an incorrect signal decision could affect road safety.

Third, the proposed architecture has not been experimentally validated in this sample paper.

Finally, real-world deployment would require appropriate hardware, traffic-controller integration, safety testing, and regulatory approval.

XIII. FUTURE WORK

Future work can include the development of a prototype using a camera and a local computing device.

A machine-vision model can be evaluated for vehicle detection and traffic-state estimation. Emergency vehicle detection can also be investigated.

The system can initially be evaluated using a traffic simulation environment before considering a physical intersection.

Future experiments can measure parameters such as:

- Average vehicle waiting time.
- Queue length.
- Traffic throughput.
- Emergency vehicle delay.
- Processing time.

These measurements should be obtained from actual experiments and should not be assumed in advance.

XIV. CONCLUSION

This paper presented a conceptual architecture for smart traffic lights using local AI processing. The proposed architecture uses camera-based traffic information, local processing, and a traffic-light controller to support adaptive signal decisions.

Previous research has investigated both learning-based traffic signal control and machine-vision-based adaptive traffic management [1], [2]. Building on these research directions,

the proposed architecture considers local processing and emergency vehicle priority as possible design features.

No experimental performance values are claimed in this paper. Implementation, testing, and quantitative evaluation are required as future work.

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